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Sugar on Fedora 15, the Spice for Children

One of Fedora's spin-offs is Sugar On A Stick (SOAS), which can be installed as an additional desktop environment. It's unlikely that LFY readers will use Sugar as their working environment; but if you ever have children visiting, this is a great way to expose them to the joys of Linux, and also keep them entertained.

My very limited experience has shown me that you do not need to provide any assistance to children when they are getting started with Sugar. You do not need to be able to teach a child the Sugar activities. Children just seem to know what to do—if the adults just leave them alone!

Applications in the Sugar environment are called *activities*. In this article, you will learn how to set up the Sugar desktop for the children around you from the perspective of a 'normal' Linux user.

Adding the Sugar desktop

One of the groups in the Fedora repository is the 'Sugar Desktop Environment'. Hence, all you need to do is run:

```
$ sudo yum groupinstall 'Sugar Desktop Environment'
```

This installs the core Sugar environment and a number of Sugar activities. There are a number of additional Sugar activities packaged as RPMs, the names of which are prefixed with 'sugar-'. Since you may want to set up Sugar for children of various ages and let them decide what they like, it is useful to install these additional activities as well. This is easily accomplished with the following command:

```
$ sudo yum install 'sugar-*
```

Sugar activities are written in Python. Sugar-specific data and activities are installed in `/usr/share/sugar`. The activities are in the activities sub-directory. There will be a `sugar.desktop` file in `/usr/share/xsessions`, so that users can opt to run the Sugar desktop from the GDM login screen. You can also try Sugar using the `sugar-emulator` command, which will run the Sugar desktop in a Xephyr window.

A user's runtime Sugar environment

The user's configuration files, log files, and other runtime information of the Sugar environment is stored in the `$HOME/.sugar` directory. To diagnose any problem with an activity, check the log files in `$HOME/.sugar/default/logs`. These are recreated for each session, and do not grow large. On the other hand, you cannot view the log of a previous session.

There is a *Log activity* to view logs from within the Sugar environment, which may not be visible on the main screen (which shows you your favourite activities in a circle or a spiral). Selecting your favourite activities is quite simple. The 'List View' icon on the upper left of the first page shows you all the activities available, and lets you 'star' the preferred applications. As you can imagine, this is very useful, as at the start, you can customise the menu to show only the appropriate or useful activities to a child.

You are not restricted to activities packaged by Fedora. Many more activities are available at <http://activities.sugarlabs.org> and packaged as a XO file, which is just an archive or a bundle, as Sugar likes to call it. You can download a bundle and install it using the command `sugar-install-bundle`. Activities installed by a user are put in the `$HOME/Activities` directory, and thus available only to the user who installed them.

Some of my favourite activities

My favourite activity is *Turtle*, which is a visual implementation of Logo. It is great to build programs by plugging Logo-like blocks and watching the turtle move. A number of samples are included to show you what is possible.

The *Espeak* activity is fun. Hindi is one of the languages included in the text-to-speech engine. The voice is pretty robotic, but I am sure it is still a pretty good input for children to learn reasonably